The Host Is Part of the Block: A Goal-Pursuit Ladder at Two Scales and the Composition-Matching Law

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Abstract

The cadence notes [1, 2] left a goal living in a conditioning channel and a creature-scale loop that holds it; this note makes the goal DO something. A registered ladder (G0-G3 plus a temporal-pressure rung) puts a frozen host on a chained-clue errand — look, open, submit, under a tick budget, with every episode's actions and observations recorded as first-class transcripts — at 230M with the validated activation-packet channel and at 27B with the validated weight-block channel. Three findings. First, the scale answer: zero-shot, BOTH hosts are below the task (0.00 with the goal in text); trained by 384 steps of oracle cloning, both are perfect when the goal rides in text (1.00 at every seed, both venues); trained with the goal in the channel, both carry a partial want (0.70/0.80/1.00 at 230M, 0.60/0.80/0.70 at 27B) that directs behavior without surviving symbol-level conflict — the interesting gap is channel-vs-text, not small-vs-large. Second, the environment fought the preregistration three times — a reveal-order leak, a mis-specified control floor, a Markovian degeneracy — and the registered controls caught all three; the diagnosis runs are kept. Third, and the note's title: a conditioning artifact works exactly on the host composition it was trained under. The same emitter that passes its gate in-harness passes through the production server's cached-generate delivery path on a matched host (0.60 vs floors 0.20/0.00) and collapses to chance when one more block arms (0.20), because the harness composes blocks by addition and the server by factor averaging — a cross-term operator mismatch independently documented in the serving stack's own audit. The law held four times in one arc (two GT rungs, two canaries), and its constructive corner stands: packets retrained under the moving host keep their write-once regime at 0.97–1.00, and the trained errand policy delivered through the real serving path reaches 0.40 against a 1.00 harness ceiling and a 0.00 serve floor — the transfer gap is now a measured number, not a fear.

1 From a want that persists to a want that acts

Paper 12 measured a goal packet that endures; paper 13 closed its re-encoding gap. The registered goal-pursuit line asks the next question: can the want DRIVE something — a multi-step errand through tools — and where does the want have to live for that to work? The environment is deliberately small and fully machine-validated: eight places, eight items, three-hop clue chains (a note names the next place, the last place holds the target beside a generator-drawn distractor, submit ends the episode), a tick budget of eight, greedy decode, and a transcript for every episode. The distractor makes the final submit a forced choice only the goal can decide; the oracle needs four ticks. Venues: the 230M creature stack (dream4 armed, the activation-packet channel the

venue	trained text	goal block	block done-choice	no-goal floor
230M (activation packet)	1.00/1.00/1.00	0.70/0.80/1.00	0.70/0.89/1.00	0.60/0.80/0.70
27B (weight block)	1.00/1.00/1.00	0.60/0.80/0.70	0.60/0.80/0.70	0.50/0.50/0.80

Table 1: G1 at both venues, per seed. Both channels carry a partial want; text carries a whole one; the floors are the trained goal-free bridges whose submit is a coin flip.

C-line validated) and the bare 27B column (the weight channel the multi-sensor line validated) — per-venue channels, registered as such, because the claim is about WHERE the want lives, not one mechanism.

2 The environment fought back, and the controls won

Three times the preregistered design was wrong, and three times its own control cells said so before any claim shipped. The first G1 run scored 1.00 everywhere INCLUDING the goal-free floor whose submit is a coin flip by construction — the reveal listed the target first, and every trained arm had learned "submit the first-listed item." The floor cell and the shuffled control caught it; the order is now randomized; the leak run is kept as the diagnosis. The shuffled check itself was then found mis-specified (registered against an untrained floor any trained policy beats) and re-registered against the trained goal-free floor. And G2's truncation rung returned its registered reading (iii): each clue names the next move, so the errand is Markovian and last-observation context recovers full performance exactly — the rung cannot separate state carriers on this task shape, recorded as a task-design note. Its one live cell: a goal entering the packet once and surviving on-policy re-encoding lands at 0.60/0.50/0.60 — direction above chance, weaker than text.

3 The scale answer

Table 1 and Figure 1 (left) carry it. Zero-shot, the errand defeats both hosts — 0.00 success with the goal stated in text, at 230M and at 27B alike (the 27B's transcripts do not even wander well). Trained by oracle cloning with the goal in text, both hosts are perfect: 1.00 at every seed, both venues. Trained with the goal ONLY in the channel, both venues land in the same band — 0.70/0.80/1.00 at 230M through the activation packet, 0.60/0.80/0.70 at 27B through the weight block — above their trained goal-free floors, below their text ceilings, and in packet-vs-context conflict cells the channel's item is almost never what gets submitted. The channel carries direction, not symbols, at both scales; scale bought nothing on the channel; and the question the program inherits is not "is the model big enough" but "what does a channel have to be for a want to survive contact with language."

4 The composition-matching law

Four independent cells, one law: *a conditioning artifact works on the host composition it was trained under, and on no other.* (1) GT at 230M: a policy bridge trained on the dream4-only host scores 1.00 — and 0.00 the moment the production temporal block's delta is summed underneath it at evaluation (none/countdown/temporal = 1.00/1.00/1.00 / 1.00/1.00/1.00 / 0.00/0.00/0.00). (2) GT at 27B reproduces the shape (0.00/1.00/0.00 on the temporal arm). (3) The serving canary:

venue	none	countdown	temporal block
230M	1.00/1.00/1.00	1.00/1.00/1.00	0.00/0.00/0.00
27B	1.00/1.00/1.00	1.00/1.00/1.00	0.00/1.00/0.00

Table 2: GT per seed: the text arms are ceilinged (the trained policy is oracle-optimal, so the tight budget exerts no pressure) and the temporal arm measures host mismatch, not time-feeling — the bridge never trained under the summed host.

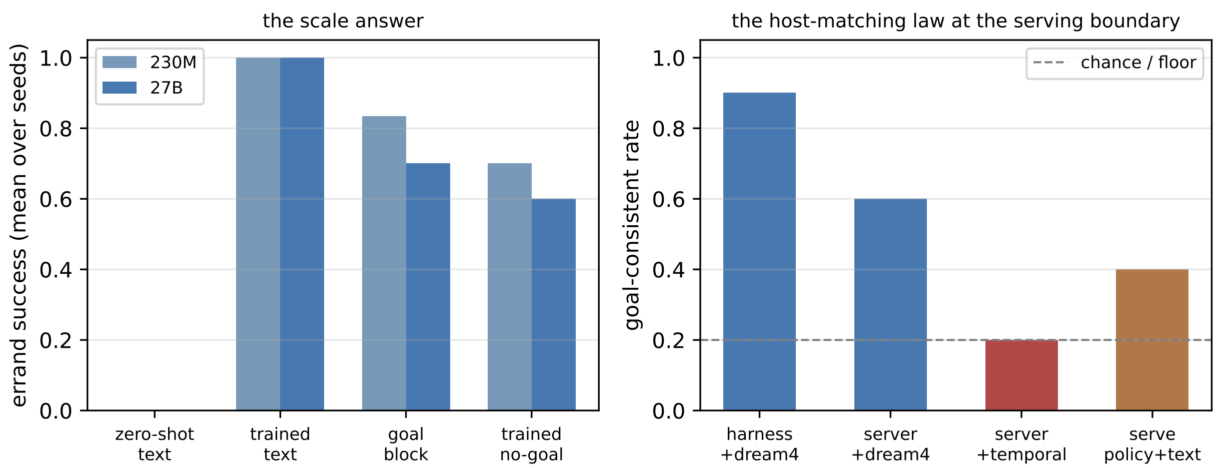

Figure 1: Left: the scale answer — zero-shot below task, trained-text perfect, channel partial, at both venues. Right: the law at the serving boundary — the same artifacts pass on matched hosts (harness+dream4, server+dream4), collapse under one extra block (server+temporal, the operator mismatch), and the trained policy delivered through the real serving path reaches 0.40 against a 1.00 harness ceiling.

the same emitter that passes its C1 gate in-harness passes through the production server's cached-generate delivery path on a matched dream4-only host — true 0.60 against floors 0.20 and 0.00 — validating the delivery wiring end to end. (4) Arm one more block and the same canary collapses to chance (0.20), even for an emitter retrained under the moving temporal host; the residual mismatch is the OPERATOR: the harness composes blocks by flat-delta addition, the server by LoRA factor averaging with cross-terms — a bug the serving stack's own audit had independently documented. Two investigation lines, one root cause; the fix pays both. The law's constructive corner: the emitter retrained under the moving host keeps its write-once packets at 0.97–1.00 across sixteen ticks (1.00 early, 0.97 late), while its live re-encoding loop degrades (0.55/0.25) — the store architecture is robust where the self-loop is not.

5 The serving stake, measured

G3 ran the errand against an isolated instance of the production server — the real prompt wrapper, block composition, and cached-generate path, never the production port. Zero-shot serve floors: 0.00 with and without goal text; oracle 1.00 through the same plumbing. Then the trained policy bridge, exported as a packet store and delivered through the server's own generation path: **policy+goal-text 0.40** — the first positive serve-side block result in the program — and policy without goal text 0.10, confirming the division of labor (the channel carries competence, the text carries the want). The serve wrapper's cost against the 1.00 harness ceiling is now a number (0.60), with a registered follow-up: retrain the policy under the serve template — the same law, applied to the prompt regime.

6 What stands

A creature-scale stack now has the full chain measured: a want that persists (paper 12), re-writes itself (paper 13), directs a multi-step errand when trained (this note), and partially survives delivery through the real serving path — with the one blocking defect between here and the live creature located, named, and independently confirmed: the serving stack's composition operator. The law this arc kept re-measuring belongs beside papers 7 and 9's negatives: zero weight-space interference was never enough, and matching the host — its blocks, its operator, its template — is not a detail of deployment but part of the block's identity. Every registration, amendment, diagnosis run, and full episode transcript is in the tracks' PLAN files and the bundled artifacts.

7 Reproducibility

Thirteen artifacts are bundled beside this note — the goal-pursuit ladder at both venues (including the kept leak-diagnosis run), the serve-composition emitter, and both canaries; every number and the figure render from them alone. Harnesses: `goal_pursuit.py`, `goal_pursuit_27b.py` and `g3_serve.py` in `tracks/goal-pursuit/`; `own_cadence.py` and `c3b_canary.py` in `tracks/own-cadence/`; registrations and results, each dated before its run, in the tracks' `PLAN.md` files.

References

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